

The need for open source AI-based DevOps tools

AI-based DevOps tools are required to:

- Avoid or reduce vendor lock-in through open standards, portable formats and interoperable APIs.
- Reduce total cost of ownership for commodity capabilities while preserving investment for differentiated engineering capabilities.
- Accelerate innovation by leveraging large communities, reusable components and rapid project evolution.
- Support hybrid and multi-cloud architectures with portable tooling and automation.
- Increase extensibility, and integrate tools into enterprise workflows, platforms, policy engines and AI agents.
- Improve transparency and auditability through inspectable code, public issue trackers and community practices.
- Enable developer choice while providing standardised enterprise paved roads.
- Reduce duplicated capabilities by

rationalising the toolchain around common platforms and open interfaces.

Features of open source DevOps tools

Tools for release management, provisioning, configuration management, systems integration, monitoring, control, and orchestration are important for DevOps adoption. These tools should have the following features:

- **Open standards and APIs:** Strong interoperability and automation interfaces, avoiding closed integration patterns.
- **Security and supply-chain posture:** Signed releases, vulnerability response, SBOM/provenance support, and transparent security practices.
- **Enterprise operability:** High availability, backup/restore, role-based access control, audits, observability, upgrade paths and automation.
- **Extensibility:** Plugins, APIs, controllers, operators, webhooks and policy integration.

- **Licence fit:** Licences must be compatible with enterprise distribution, SaaS, modification and commercial use.
- **Portability:** Run consistently across cloud, on-premises and hybrid environments where required.
- **Sustainability:** Clear project governance, funding, maintainers and exit strategy.

Mapping open source tools to the different phases of the DevOps lifecycle

A DevOps toolchain is a combination of tools that help in the delivery, development, and management of applications throughout the system's development lifecycle. Software teams need to automate the entire cycle of build, provisioning and deployment of test environments, including the tools, scripts and test data to ensure rapid delivery. These teams need to collaborate around application architecture and monitor event-based mechanisms for seamless data flow across the toolchains.

The different stages any software or application must pass through when

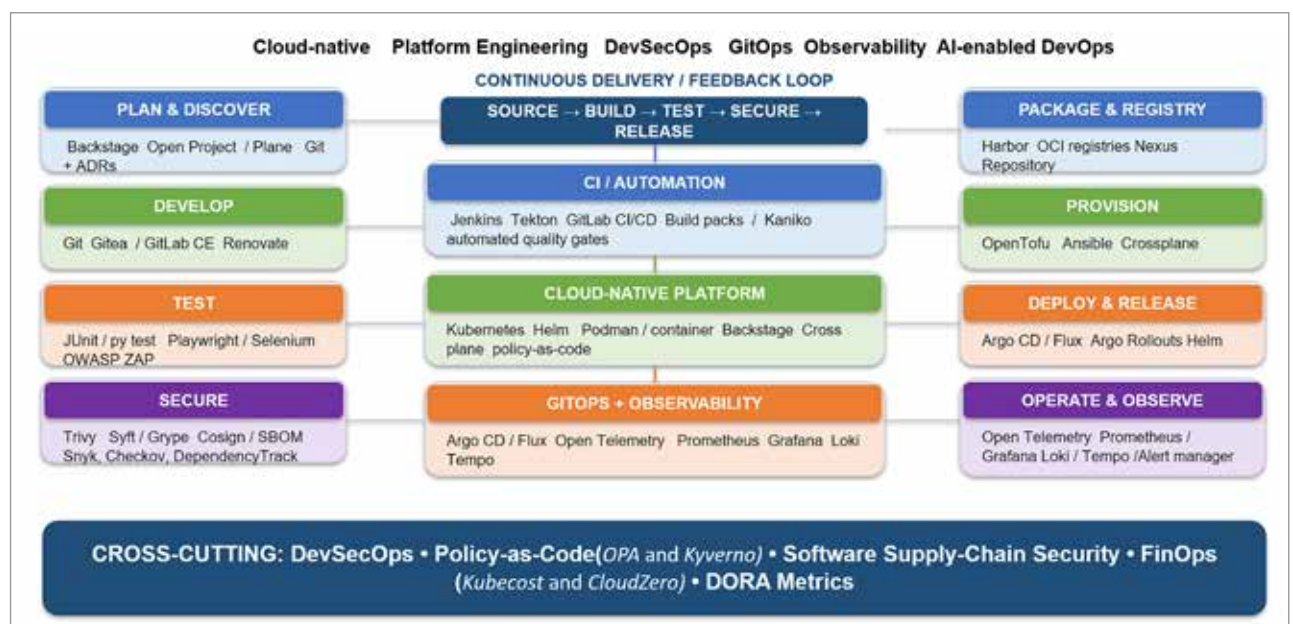


Figure 1: DevOps lifecycle and open source tools mapping